Final Design Report

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Project Description:

A high-performance professional portfolio developed using the React framework to demonstrate expertise in front-end development and system design. The platform integrates dynamic project and experience galleries, a customized UI/UX, a contact form, and cloud-based deployment. It serves as a live demonstration of full-stack principles, optimized for cross-device compatibility and streamlined professional networking.

User Interface Specification:

1. Design Philosophy

The UI follows a content-first approach. Every design choice is intended to reduce friction between the user (recruiters and peers) and the technical documentation.

Clarity: Minimalist layouts with generous white space.

Interactivity: Subtle hover states and transitions to provide tactile feedback without being distracting.

Accessibility: High contrast ratios and semantic HTML for screen-reader compatibility.

2. Visual Identity

Element	Specification
Typography	Primary: Inter or Roboto (San-serif) for high legibility. Monospace: Fira Code for technical snippets and code blocks.
Color Palette	Deep Charcoal (#121212) for backgrounds; Electric Blue (#007AFF) for primary actions; Slate Gray (#8E8E93) for secondary text.
Iconography	Minimalist line icons (e.g., FontAwesome or Lucide React) to represent different technical domains.

3. Layout Architecture

The application is structured as a Single Page Application (SPA) with the following key zones:

A. Global Navigation (Header)

Behavior: Sticky positioning to remain accessible during scrolling.

Elements: Logo (Initial-based), links to Projects, Experience, About, and a "Download Resume" CTA button.

B. About me Section

Focus: A high-impact headline and a brief sub-headline summarizing your identity as a developer/analyst.

Action: Primary button leading directly to the Featured Projects gallery.

C. Project and Experience Grid

Layout: Responsive 2 or 3-column card layout.

Card Anatomy: * Thumbnail image/Graphic representation.

Project Title and short description.

Tech Tags: Small pills indicating languages used (e.g., React, C++, Python).

D. Footer

Elements: Copyright notice, social links (LinkedIn, GitHub), downloadable resume, and an email contact form.

Test Plan and Results:

Part 1:Description

The testing strategy for my project mainly focuses on responsive design verification. Since the primary goal of the portfolio is to showcase work to potential employers, the testing ensures that all navigation links, downloadable assets (such as the Resume/CV), and contact mechanisms function correctly without requiring knowledge of the internal code structure. I will simulate user behavior across various devices (Desktop, Tablet, Mobile) to guarantee the layout adapts gracefully.

Additionally, Boundary and Negative Testing will be applied specifically to the "Contact Me" form to ensure that invalid inputs (such as missing email addresses or empty messages) are handled with clear error messages. Finally, basic Performance Testing will be conducted using browser developer tools to ensure that high-resolution project images do not negatively impact page load times, ensuring a smooth user experience.

Part 2: Test Cases

TC-01: Navigation Bar Functionality
ID: TC-01

Purpose: Verify that the main navigation bar links redirect the user to the correct sections.

Description: Click the "About Me" link or any link in the header navigation.

Inputs: User clicks on "About Me" anchor text.

Expected Outputs: The viewport scrolls smoothly to the #about section; the URL updates.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: Integration

TC-02: Projects Functionality
ID: TC-02

Purpose: Ensure the primary "View Projects" button leads users to the portfolio grid.

Description: Locate the main button in the top section and click it, as well as the carousel

Inputs: User clicks on "View Projects" button.

Expected Outputs: Page navigates/scrolls immediately to the Project Gallery section.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: Integration

TC-03: Resume Download
ID: TC-03

Purpose: Verify that the resume file is accessible and downloads correctly.

Description: Click the "Download Resume/CV" button.

Inputs: User click.

Expected Outputs: A PDF file named Firstname_Lastname_Resume.pdf opens in a new tab or initiates a download.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: System

TC-04: Contact Form - Valid Submission
ID: TC-04

Purpose: Validate that the contact form sends data when all fields are correct.

Description: Fill out the contact form with valid data and press "Send."

Inputs: Name: "Test User", Email: "test@example.com", Msg: "Hello world".

Expected Outputs: Form clears, and a "Thank you for your message" success toast/modal appears.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: Integration

TC-05: Contact Form - Empty Fields (Validation)
ID: TC-05

Purpose: Ensure the user cannot submit the form without the required data.

Description: Leave all fields blank and attempt to click the "Send" button.

Inputs: Null (Empty fields) + Click "Send".

Expected Outputs: Form is not submitted; error messages ("This field is required") appear under Name and Email fields.

Case Indication: Abnormal

Box Type: Blackbox

Test Type: Functional

Level: Unit

TC-06: Contact Form - Invalid Email Format
ID: TC-06

Purpose: Check email format validation logic.

Description: Enter an email address missing the "@" symbol or the domain.

Inputs: Email: "user.com" (missing @) + Click "Send".

Expected Outputs: Form is not submitted; browser or custom tooltip displays "Please enter a valid email address."

Case Indication: Boundary

Box Type: Blackbox

Test Type: Functional

Level: Unit

TC-07: External Social Links
ID: TC-07

Purpose: Verify that social icons link to the correct external profiles.

Description: Click the LinkedIn icon in the footer.

Inputs: User clicks on the fa-linkedin icon.

Expected Outputs: A new browser tab opens, loading the correct LinkedIn profile URL.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: System

TC-08: Project Image Loading
ID: TC-08

Purpose: Ensure project thumbnails load correctly and are not broken.

Description: Load the main portfolio page and visually inspect the project grid.

Inputs: Page Load (HTTP GET request).

Expected Outputs: All <img> tags render the correct source files; no "broken image" icons are visible.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: System

TC-09: Mobile Menu Toggle
ID: TC-09

Purpose: Verify the menu works on small screens.

Description: Resize the browser window to mobile width (<768px) and click the hamburger icon.

Inputs: Window width: 400px; Click on Menu Icon.

Expected Outputs: Navigation links slide in or drop down; the icon changes to an "X" (close) state.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: Integration

TC-10: Cross-Browser Compatibility
ID: TC-10

Purpose: Ensure the site renders correctly on a different browser engine.

Description: Open the hosted URL in Firefox (Gecko engine) instead of Chrome (Blink).

Inputs: URL entry in Firefox Address bar.

Expected Outputs: Layout remains consistent; no overlapping text or misaligned elements compared to Chrome.

Case Indication: Normal

Box Type: Blackbox

Test Type: Functional

Level: System

TC-11: 404 Page Not Found
ID: TC-11

Purpose: Verify user experience when accessing a non-existent route.

Description: Manually type a nonsense path into the URL bar.

Inputs: URL: www.yourportfolio.com/xyz-nonsense.

Expected Outputs: User is redirected to a custom 404 page with a "Back to Home" link, not a generic browser error.

Case Indication: Abnormal

Box Type: Blackbox

Test Type: Functional

Level: System

TC-12: Page Speed / Performance
ID: TC-12

Purpose: Ensure the site meets basic performance benchmarks.

Description: Run a Google Lighthouse audit on the homepage.

Inputs: Developer Tools -> Lighthouse -> "Generate Report".

Expected Outputs: Performance score is > 90; Accessibility score is > 90.

Case Indication: Normal

Box Type: Blackbox

Test Type: Performance

Level: System

Part 3: Chart

Test Case ID	Normal / Abnormal / Boundary	Blackbox / Whitebox	Functional / Performance	Unit / Integration
TC-01	Normal	Blackbox	Functional	Integration
TC-02	Normal	Blackbox	Functional	Integration
TC-03	Normal	Blackbox	Functional	Integration
TC-04	Normal	Blackbox	Functional	Integration
TC-05	Abnormal	Blackbox	Functional	Unit
TC-06	Boundary	Blackbox	Functional	Unit
TC-07	Normal	Blackbox	Functional	Integration
TC-08	Normal	Blackbox	Functional	Integration
TC-09	Normal	Blackbox	Functional	Integration
TC-10	Normal	Blackbox	Functional	Integration

TC-11	Abnormal	Blackbox	Functional	Integration
TC-12	Normal	Blackbox	Performance	Integration

Part 4: Results

ID	Test Case Title	Type/Level	Result	Manual Verification / Observation
TC-01	Nav Bar Functionality	Functional / Integration	**PASS**	All buttons work as expected
TC-02	Projects Functionality	Functional / Integration	**PASS**	All pills work, and all videos/ pictures look as expected
TC-03	Resume Download	Functional / System	**PASS**	The resume is displayed and can be clicked to be downloaded
TC-04	Contact - Valid Submission	Functional / Integration	**PASS**	Contact form correctly formats email and sends it to my email

TC-05	Contact - Empty Fields	Functional / Unit	**PASS**	Will not send with empty fields
TC-06	Contact - Invalid Email	Functional / Unit	**PASS**	Will not send with an invalid email
TC-11	404 Page Not Found	Functional / System	**PASS**	Correctly displays the error message
TC-07	External Social Links	Functional / System	**PASS**	All links work properly
TC-08	Project Image Loading	Functional / System	**PASS**	All images load in correctly with no issue
TC-09	Mobile Menu Toggle	Functional / Integration	**PASS**	Mobile view was fully re-worked to be user-friendly
TC-10	Cross-Browser	Functional / System	**PASS**	Works on any browser

TC-12	Performance (Lighthouse)	Performance / System	**PASS**	Scored in the high 70s, for my first website and a solo project, I'm happy with this

User Manual:

To best view the user manual, please click the link to my GitHub page:
User Manual
If you wish to view by PDF, I have included that as well

Personal Portfolio: User Guide & Manual

Version: 1.0 **Date:** February 5, 2026

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Overview

This Personal Portfolio website serves as a centralized hub to showcase professional skills, software engineering projects, and technical experience. It is designed for recruiters, potential clients, and peer developers to easily access resume details and contact information.

The application is a Single Page Application (SPA) built using **React**, ensuring a responsive and seamless user experience.

Installation Guide

This section is intended for developers or graders who wish to run the portfolio source code locally.

Prerequisites

Before you begin, ensure you have the following installed on your machine: * **Node.js** (v14.0.0 or higher) * **npm** (normally comes with Node.js) * A code editor (e.g., VS Code)

Setup & Deployment

1. **Clone the Repository** Open your terminal and clone the project: `bash` `git clone [https://github.com/your-username/personal-portfolio.git](https://github.com/` `cd personal-portfolio`
2. **Install Dependencies** Install the necessary React dependencies listed in `package.json`: `bash` `npm install`

3. **Start the Application** Run the app in development mode: `bash npm start` Open http://localhost:3000 to view it in your browser. The page will reload when you make edits.

User Manual

This section explains how to interact with the live website. The interface is divided into key sections: **Home**, **Skills**, **Projects**, and **Contact**.

Navigating the Site

The **NavBar** is fixed at the top of the screen, allowing you to jump quickly between different sections without reloading the page.

- **Home:** Returns you to the introductory Banner.
- **Skills:** Jumps to the technical proficiency section.
- **Projects:** Displays the project gallery.
- **Contact:** Scrolls to the inquiry form.

> **Note:** On mobile devices, the NavBar may collapse into a hamburger menu icon. Tap the icon to expand the navigation options.

The Banner (Home)

Upon loading the site, you are greeted by the **Banner**. This section introduces the developer and provides a brief professional summary. * **Action:** Click the "My Work" button (if available) to scroll down to the Projects section immediately.

Viewing Skills

The **Skills** component categorizes technical abilities (e.g., Languages, Frameworks, Tools). * Hover over specific skill icons to see the name of the technology.

Browsing Projects

The **Projects** section displays a gallery of completed work. 1. **Project Cards:** Each project is displayed as a card containing a title, description, and technology stack. 2. **Learn More:** Click on a project card or the "View Code" link to be taken to the specific GitHub repository or live demo for that project.

Contacting the Developer

The **Contact** section allows you to send a direct message. 1. Enter your **Name**, **Email**, and **Message** in the provided fields. 2. Click **Submit**. 3. You should see a confirmation alert that your message has been sent.

Privacy Note: Your email address is only used to reply to your inquiry and is not stored publicly.

FAQ

Q: Is the portfolio mobile-friendly? A: Yes. The application is built with responsive CSS, ensuring it looks good on desktops, tablets, and mobile phones.

Q: Where can I find a downloadable version of the resume? A: A "Download Resume" button is located in the Banner section (or footer), which will provide a PDF version of the full CV.

Q: I found a bug in the layout. How do I report it? A: Please submit an issue by using the Contact form on the website to describe the problem.

Q: Why are images taking a moment to load? A: High-quality project screenshots may take a moment to load depending on your internet connection. We use lazy loading to ensure the rest of the site remains interactive while images fetch.

Spring Presentation:

To best view my Spring Presentation, click the link below to my GitHub, where you can better view the poster
[Spring PowerPoint]

Expo Poster:

To best view the expo poster, please click the link below:
[Expo Poster]
Otherwise, I have included an image below

Fall Self-Assessment:

I have included a link to my GitHub page to view my fall self-assessment below:
Fall Self-Assessment

Spring Self-Assessment:

I have included a link to my GitHub page to view my spring self-assessment below:
Spring Self-Assessment

Summary of Hours and Justification:

I have included a link to my GitHub page to view my summary of hours worked below:
Summary of Hours

The only thing that might have taken longer than I recorded would be developing the HTML structure for the website. I would say that it took around 10 hours. While this project might have taken someone with web development experience much less time, as this was a large learning experience, it took me around 55 hours to fully complete. If I remade the website from scratch, I could probably do it in less than half that time, but learning the framework added a lot of time.

Counting the fall semester, I spent about 110 hours working on researching, developing, and testing this project. As I worked alone, all work seen was done by me; I feel the hours are justified due to the amount of research and the long development cycle, since this was my first time doing a full-blown front-end project where I had lots to learn.

Summary of Expenses:

- All software and hosting services: free
- Domain name: $6
- Expo poster materials: $30

Appendix:

As I worked alone on this project, there were no meeting notes to include. I have justified the hours worked on this project above. I do not have any external GitHub repositories I pulled from or any major references or citations.